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# The impact of ESG factors on credit ratings: An empirical study of European banks

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A R T I C L E I N F O

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A B S T R A C T

This paper explores the impacts of ESG factors on credit ratings assigned by Moody's, S&P, and Fitch to 106 European listed banks over the period from 2019 to 2023. The findings reveal that environmental factors exhibit a positive and significant influence on credit ratings from all three agencies, highlighting the integration of ESG factors into credit rating methodologies by major rating agencies. However, ESG Combined Score along with social, governance and controversial factors yield mixed results, indicating complexities in their measurement and their impact on creditworthiness of banks. The study emphasizes the critical need for standardized frameworks and enhanced transparency in ESG reporting to strengthen the integration of ESG factors into credit risk assessments of financial institutions.

## 1. Introduction

ESG risks—environmental, social, and governance factors—can adversely affect the creditworthiness of financial institutions. These risks include environmental exposures, social controversies, and governance deficiencies, potentially leading to reputational damage, regulatory scrutiny, and higher funding costs for both banks and non-bank financial institutions (NBFIs). Climate-related risks, such as physical and transition risks, can reduce the value of loan portfolios and collateral. Banks involved in controversial lending, such as financing projects with negative social impacts or human rights violations, may face reputational harm, legal liabilities, and customer attrition. Governance failures can weaken the credibility of financial reporting, internal controls, and ethical standards, resulting in penalties, lawsuits, and loss of investor confidence. Integrating ESG factors into credit rating methodologies has become a critical focus of research and practice in the financial sector (see Tables 1 and 2).

Recognizing the adverse effects of ESG risks on creditworthiness, banks are increasingly integrating environmental, social, and governance considerations into their practices. This shift includes greater participation in green financing and a heightened focus on ESG performance to tackle environmental challenges and promote sustainable development. As global concern over climate change grows, banks are adapting by transitioning from traditional models to offering eco-friendly products, aligning with emerging market trends (Dikau & Volz, 2021).

The regulation of ESG factors has significantly evolved in recent years, particularly within the European banking sector. Although formal regulations were initially limited, the early 2000s saw growing recognition of the importance of ESG in financial decision-making. A major turning point came with the launch of the Principles for.

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The regulations of ESG has undergone significant revolutions in recent years, particularly within the European banking sector. In the early 2000s, formal ESG regulations were limited, but the importance of ESG in financial decision-making began to gain recognition. A major milestone came with the launch of the Principles for Responsible Investment (PRI) in 2006, which provided a framework for integrating ESG considerations across asset classes. In 2017, the Task Force on Climate-related Financial Disclosures (TCFD) emerged as a pivotal framework, promoting transparency in climate-related disclosures. TCFD's recommendations guide organizations in reporting governance, strategy, risk management, and metrics, enabling better assessment of climate-related risks. By September 2023, 312 banks endorsed the TCFD framework, up from 180 in October 2021 and 92 in March 2020, according to Moody's Investor Service. The European Union (EU) advanced sustainable finance with the 2018 EU Sustainable Finance Action Plan, aiming to redirect capital flows towards sustainable investments, integrate sustainability into risk management, and enhance financial market transparency. Key initiatives include the EU Taxonomy Regulation, which classifies environmentally sustainable activities, and the Sustainable Finance Disclosure Regulation (SFDR), which mandates ESG-related disclosures by financial market participants. In response to growing demand for climate risk reporting, banks joined voluntary frameworks like the Net Zero Banking Alliance (NZBA), launched in 2021. This UN-organized consortium of banks aims to reduce global greenhouse gas emissions to zero by 2050. By May 2024, 144 banks from 44 countries, with nearly half based in Europe, had joined NZBA. These developments, alongside national and industry initiatives, have pushed European banks towards greater ESG integration and shaped the future of sustainable finance in the region.

Building on these advancements, the Corporate Sustainability Reporting Directive (CSRD), officially adopted by the European Union Council and entering into force on January 5, 2023, represents a significant expansion of ESG reporting requirements. The CSRD extends the scope and the reporting requirements of the Non-Financial Reporting Directive (NFRD), which previously applied only to companies with over 500 employees. Under CSRD, large companies — defined as those with more than 250 employees, EUR50 million in turnover, and EUR25 million in total assets — are now required to apply. Additionally, all listed companies (excluding micro-

**Table 1**
Related literature on ESG and performance in the banking sector.

| Variable | Reference | Topic | Findings |
|---|---|---|---|
| ESG, E, S, G | Buallay (2018) | Relationship between ESG and bank's operational (ROA), financial (ROE) and market performance (Tobin's Q) over 2017–2016 | Significant positive impact of ESG on the performance, the relationship between ESG disclosures varies if measured individually |
| ESG | Menicucci and Paolucci (2022) | Impact of ESG scores on bank performance in the Italian banking sector over 2016–2020 | ESG policies negatively affect operational and market performance in the banking sector |
| ESG | Yuen et al. (2022) | Impacts of the environment, social and governance (ESG) and its components on global bank profitability from 2006 to 2021 | ESG activities may reduce bank profitability, thus supporting the trade-off hypothesis that adopting ESG standards could increase bank costs while lowering profitability |
| ESG | Ng et al. (2020) | Connection between financial development and ESG performance in Asiaover 2013–2017 | Financial development is positively related to ESG success |
| E, S, G | Bătae et al. (2021) | Relationships between ten dimensions of the environmental, social, and governance pillars and bank financial performance over 2010–2019 | Positive relationship between emission reductions and financial performance, negative relationship between social responsibility policies and market performance, negative relationship between corporate governance and financial performance |
| E, S, G | Shakil et al. (2019) | Effect of ESG scores on bank performance in emerging markets over 2015–2018 | Positive association of emerging market banks' environmental and social performance with financial performance, governance has no influence |
| E,S | Nizam et al. (2019) | Impact of access to finance and environmental financing on the financial performance of the banking sector globally over 2013–2015 | Access to finance has significantly positive effects on banks' financial performance |
| S | Wu and Shen (2013) | Association between corporate social responsibility (CSR) and financial performance, and discusses the driving motives of banks to engage in CSR over 2003–2009 | CSR positively associates with FP in terms of return on assets, return on equity, net interest income, and non-interest income. CSR negatively associates with non-performing loans. |
| S | Siueia et al. (2019) | Impact of voluntary CSR disclosure on Financial Performance (FP) in the Sub-Saharan banking sector between 2012 and 2016 | Significant and positive relationship between FP and CSR disclosure, suggesting that CSR behavior is helpful to improve the performance of banks. |
| G | Zagorchev and Gao (2015) | How corporate governance affects financial institutions in the U.S. between 2002 and 2009. | Better governance is negatively related to excessive risk-taking and positively related to the performance of U.S. financial institutions. |
| G | Sohailet al. (2017) | Explore how governance mechanisms (internal and external) enhance the performance of the return on asset (ROA), return one equity (ROE), earning per share (EPS) and dividend payout ratios (DP) of the banks of Pakistan over 2018–2014 | Internal and external governance enhance the performance of the banking sector of Pakistan |
| G | Maxfield and Wang (2022) | impact of bank corporate governance reforms over 2004–2011 | The post-crisis corporate governance reforms in the banking sector appear to be effective in promoting greater bank attention to non-shareholder stakeholders' interests |

enterprises) must adhere to the directive. Approximately 50,000 companies across the EU, representing 75 % of all EU companies' turnover, will be subject to detailed EU sustainability standards. Non-European companies with EU branches or subsidiaries generating a net turnover of EUR150 million or more within the EU will also need to adhere to the CSRD. The CSRD marks a significant step forward in standardizing and elevating sustainability reporting across Europe, introducing mandatory standards and enhanced quality assurance requirements. Companies must prepare for their first reports in 2025, covering fiscal years starting on or after January 1, 2024, and align with the European Sustainability Reporting Standards (ESRS). These disclosures will be integrated into the management report of annual reports and initially subject to limited assurance by statutory auditors.

As ESG factors become increasingly important in financial decision-making, stakeholders in the financial industry are recognizing the need to integrate these considerations into credit risk assessment and rating practices. This shift is evident in initiatives like the Principles for Responsible Investment (PRI), which issued a 2016 statement on ESG in credit risk and rating, backed by over 180 investors and 28 credit rating agencies, including major players like S&P Global Ratings, Moody's Corporation, and Fitch Group. The statement underscores the necessity for systematic and transparent integration of ESG factors into credit ratings and financial analysis. Today, all three major credit rating agencies explicitly incorporate ESG risks and opportunities into their methodologies. For instance, S&P Global acknowledges the increasing demand for transparency regarding how environmental, social, and governance factors are factored into its credit rating processes. From July 2015 to August 2017, environmental and climate-related concerns influenced corporate ratings in 717 cases, with 106 leading to changes such as upgrades, downgrades, or CreditWatch placements. Moody's formalized its ESG integration approach in January 2019 through its ESG General Principles Methodology, introducing tools like the Credit Impact Score (CIS), Issuer Profile Score (IPS), and Carbon Transition Indicators (CTIs). Fitch developed its proprietary ESG Relevance Scores, showing how ESG factors affect credit rating decisions. These advancements mark a significant step in aligning credit risk assessment with the broader goals of sustainable finance and responsible investing.

However, amid current political concerns such as economic downturns, inflation, and regional conflicts, there is a growing perception that ESG considerations may be raising costs and hindering investment in certain sectors. A white paper by Fitch revealed that some corporate leaders are already contemplating postponing ESG projects in preparation for potential recessionary challenges. Among 1325 CEOs surveyed from companies with annual revenues of at least USD 500 million, nearly half expressed intentions to "pause or reconsider" their ESG efforts in the next six months, with around 34 % having already taken such actions. S&P also reported a 44 % decline in ESG-related credit rating actions in 2023 compared to 2022, largely due to the diminished impact of COVID-19-related risks.

Given the current economic uncertainties and shifting market dynamics, it has become increasingly crucial to evaluate how these changes are affecting the financial sector, especially the bond market, which represents the largest asset class in Europe. Recent trends show a growing emphasis by credit rating agencies on incorporating ESG factors into their methodologies, raising questions about the effect on bank credit ratings, given their central role in debt issuance. Data from the European Securities and Markets Authority (ESMA) reveals that covered bonds have substantial trading volumes, second only to sovereign bonds. In this context, it is essential to examine the correlation between banks' ESG performance and their credit ratings, along with the key ESG sub-factors prioritized by rating agencies and any differences in their methodologies. This analysis will offer valuable insights into the relationship between ESG factors and creditworthiness in the banking sector, providing issuers with critical perspectives on the risks and opportunities linked to sustainable finance practices.

This paper analyzes the impact of the ESG Combined Score, Environmental Pillar Score, Social Pillar Score, Corporate Governance

**Table 2**
Related literature on ESG and credit risk in the banking sector.

| Variable | Reference | Focus | Findings |
|---|---|---|---|
| ESG | Tommaso & Thornton, (2020) | Whether environmental, social and governance (ESG) scores of European banks impact on their risk- taking behavior and bank value. | High ESG scores are associated with a modest reduction in risk-taking for banks that are high or low risk-takers, and the impact is conditional on executive board characteristics. |
| ESG, E, S, G | Liu et al. (2023) | Association between a bank's nonperforming loans and its ESG performance from U.S. commercial banks | Banks' ESG ratings are negatively associated with their nonperforming loans. Banks' high performance in all three pillars of ESG ratings reduces their nonperforming loans. |
| ESG,E, S, G, C | Bruno et al. (2023) | Impact of ESG COM scores on non-performing loans for European listed banks over 2002–2020 | Banks with greater levels of the ESG score have higher levels of NPLs. |
| ESG | Tóth et al. (2021) | Relationship between financial stability and ESG performance of 243 banks from the European Union and the European Free Trade Association | ESG performance reduce the ratio of non- performing loans significantly |
| E, S, G | Devalle et al. (2017) | How ESG performance influences credit ratings of Italian and Spanish public companies in the fiscal year 2015 | ESG performance is positively associated with higher credit ratings |
| C | Samaniego-Medina & Giráldez-Puig (2022) | Effect of ESG controversies on the credit rating of the European banking sector, involving 65 European banks from 18 countries in the 2011–2020 period | The lower the level of ESG controversies, the greater the probability of achieving the highest credit ratings. |

Pillar Score, and ESG Controversies Score, obtained from the Refinitiv database, on the credit ratings of European banks. We chose European banks for this study due to their increased vulnerability to ESG risks, driven by the region's stringent regulatory environment and high expectations for sustainability and corporate governance.

To achieve our objectives, we used a panel dataset comprising 106 European banks from 25 countries, with credit ratings available for the period from 2019 to 2023. The analysis was conducted using the statistical software Stata, applying ordered logistic regression models to assess the influence of five ESG indicators on the credit ratings assigned by the three major international rating agencies: S&P, Moody's, and Fitch.

Our research reveals a significant positive correlation between the ESG Combined Score and Fitch's credit ratings, indicating that Fitch places greater weight on overall ESG performance. Environmental factors consistently showed a positive impact on credit ratings across Moody's, S&P, and Fitch, underscoring the financial sector's acknowledgment of climate risks and the importance of strong environmental practices in assessing creditworthiness. However, the Social Pillar Score produced mixed results: S&P's ratings showed a significant negative correlation, while Moody's and Fitch found no statistically significant relationship, reflecting the difficulties in quantifying the financial impact of social initiatives. The Corporate Governance Pillar Score had a positive effect on Fitch's ratings but no significant influence on Moody's or S&P's, highlighting differences in rating methodologies. The ESG Controversies Score significantly boosted S&P's credit ratings, emphasizing its focus on transparency and risk management. Control variables like profitability, capital adequacy, and size consistently showed positive correlations with credit ratings, while the non-performing loan ratio was negatively correlated.

This study makes several key contributions to current research. First, it builds on previous research by examining the relationship between ESG performance and credit ratings, providing fresh insights into how ESG factors influence the financial performance of banks, particularly in terms of creditworthiness. Second, the findings offer practical guidance for financial institutions by illustrating the impact of ESG performance on credit ratings, helping banks refine their strategies accordingly. Third, the research provides actionable insights for regulators by highlighting the link between ESG performance and credit ratings, which policymakers can use to craft regulations that promote stronger ESG practices while ensuring financial stability. Additionally, the study underscores the importance of integrating ESG considerations into credit rating processes to improve transparency and accountability in the financial sector. Fourth, the study offers a detailed analysis of the three ESG Pillar Scores, the ESG Controversies Score, and the Combined ESG Score, both individually and collectively. This comprehensive approach enhances the robustness of the findings and drives methodological innovation. Lastly, by including three of the world's largest credit rating agencies, the study goes beyond prior research that often focused on a single agency or specific markets. By comparing results across these agencies, the study provides a more diverse and comprehensive perspective on how ESG factors influence credit ratings, increasing the relevance and applicability of the findings for a broader audience.

The structure of the paper is as follows: Section 2 provides a review of the relevant literature. Section 3 presents the development of hypotheses. Section 4 defines the data sample, outlines the study process, and details the statistical techniques and model variables. Section 5 presents and explains the robustness tests and empirical results. Finally, Section 6 concludes the paper by discussing its limitations, implications, and offering recommendations for future research.

## 2. Literature review

### *2.1. The impact of ESG factors on credit ratings of non-financial companies*

The adoption of the CSRD marks a transformative shift in sustainability reporting, highlighting the growing importance of ESG factors in assessing a company's long-term financial health and creditworthiness. As ESG risks—such as climate change, regulatory changes, and social governance—increasingly influence business operations and profitability, understanding their impact on credit ratings becomes critical. Studying the impact of ESG factors is essential to provide insights into how enhanced ESG disclosures affect credit risk assessments, enabling investors, regulators, and companies to make more informed decisions and better manage emerging risks in a rapidly evolving regulatory landscape.

Two contrasting theories exist regarding the impact of a company's ESG investments on its financial performance. The stakeholder theory, as proposed by Freeman (2023), argues that a firm's long-term success depends on its ability to manage relationships with all stakeholders, not just shareholders—including individuals or groups affected by or interested in the firm's actions, such as suppliers, customers, employees, and the broader community. Consequently, ESG-focused businesses are expected to enhance both firm value and financial performance while improving stakeholder satisfaction (Di Tommaso & Thornton, 2020). In line with Freeman's theory, Peng and Isa (2020) have demonstrated that ESG performance, both as a whole and in its individual dimensions, is positively correlated with business performance. Some scholars have focused on specific sustainability factors. For instance, Tan et al. (2017) examined the impact of environmental performance on firm outcomes within the travel and tourism industry, finding that slack resources amplified the positive influence of environmental performance on financial performance. Similarly, Jo and Harjoto (2011) explored the causal relationship between corporate social responsibility (CSR) and corporate governance, finding that CSR participation positively affects firm value, aligning with the conflict-resolution hypothesis derived from the stakeholder theory. Furthermore, research has shown that ESG performance not only improves corporate financial performance but also that financial development enhances sustainability at both country and institutional levels (Ng et al., 2020).

On the other hand, the agency hypothesis (Jensen & Meckling, 1976) suggests that ESG-related activities can create conflicts between shareholders and managers, arguing that investing in ESG matters is not ideal for shareholders, as it depletes capital and reduces profitability. In line with the agency hypothesis, Brown et al. (2006) argued that agency costs significantly contribute to corporate giving, noting that larger boards of directors are associated with higher levels of charitable donations and the establishment of corporate foundations. Similarly, Barnea and Rubin (2010) found that higher CSR ratings are linked to increased CSR spending, further supporting the agency hypothesis.

As credit rating agencies increasingly integrate ESG factors into their assessments, recent research highlights the need to reevaluate the role of ESG in credit risk assessments. Miralles-Quiros et al. (2017) demonstrated that ESG factors play a significant role in risk assessment, while Zeidan et al. (2015) emphasized their relevance in credit policies. Devalle et al. (2017), using ordered logistic regression on a sample of 56 public companies in Italy and Spain, found that ESG performance is positively associated with higher credit ratings. Similarly, Chodnicka-Jaworska (2021) found that ESG risks significantly influence credit ratings, with environmental factors being the most impactful. Sectors like energy, industrials, materials, and utilities are most affected due to regulations on pollution and resource conservation.

### 2.2. The impact of ESG factors on credit ratings of banks

While scholars have increasingly examined the link between ESG performance and corporate financial performance, there is a notable lack of research on its impact within the banking sector. This study aims to address this gap by investigating whether a bank's ESG performance influences its likelihood of receiving higher credit ratings from major rating agencies. Unlike prior research, which has primarily focused on the impact of sustainability factors on profitability or stock returns, this study specifically explores how ESG performance influences the creditworthiness of banks, offering new insights into the role of ESG in financial stability and risk assessment.

The relationship between ESG factors and financial performance in the banking sector has been widely studied, with mixed results. Buallay (2019) found a significant positive impact of ESG disclosures on banks' financial performance. Similarly, Ng et al. (2020) highlighted a positive connection between financial development and ESG performance in Asia. Bătae et al. (2021) explored ten dimensions of ESG pillars and found a positive relationship between emission reductions and financial performance but noted negative impacts from social responsibility policies and corporate governance on market and financial performance, respectively. Conversely, Menicucci and Paolucci (2022) observed that ESG policies negatively affected operational and market performance in the Italian banking sector. Similarly, Yuen et al. (2022) supported the trade-off hypothesis, showing that ESG activities reduced global bank profitability, as adopting ESG standards increased costs.

Shakil et al. (2019) found that environmental and social performance positively influenced financial performance in emerging markets, while governance had no significant effect. Focusing on social factors, Wu and Shen (2013) found that CSR positively correlated with financial performance metrics like ROA, ROE, and income, while reducing non-performing loans. Siueia et al. (2019) reinforced this by showing a significant positive relationship between CSR disclosure and financial performance in Sub-Saharan banks.

Regarding governance, Zagorchev and Gao (2015) found that better corporate governance reduced excessive risk-taking and improved performance in U.S. financial institutions. Sohail et al. (2017) confirmed that internal and external governance mechanisms enhanced ROA, ROE, EPS, and dividend payouts in Pakistani banks. Maxfield and Wang (2022) noted that post-crisis governance reforms effectively promoted greater attention to non-shareholder stakeholders' interests in the banking sector. On the other hand, Aslam and Haron (2020) studied 129 Islamic banks and discovered that larger board sizes and the presence of risk management committees had a negative and significant impact on the performance of these banks.

The COVID-19 pandemic has underscored that a bank's creditworthiness is influenced not only by financial factors but also by non-financial factors, particularly ESG performance. During the pandemic, banks with strong ESG practices demonstrate greater resilience and financial stability, earning higher levels of trust and confidence from stakeholders, including investors, customers, and regulators.

Di Tommaso and Thornton (2020) found that high ESG ratings among European banks were associated with a modest reduction in risk-taking, regardless of the banks' initial risk levels. Research by Tóth et al. (2021), Liu et al. (2023), and Bruno et al. (2023) explored the correlation between ESG scores and credit risk. While Tóth et al. (2021) and Liu et al. (2023) identified a negative relationship between ESG scores and ratio of non-performing loan, Bruno et al. (2023) reached contradictory results, likely due to inconsistencies in ESG data. ESG ratings from different providers diverge substantially, as previously shown in Chatterji et al. (2016).

Given the connection between ESG ratings and credit risk, it is crucial to further examine how ESG performance impacts credit ratings, which serve as key indicators of a bank's overall creditworthiness. Credit ratings provide a comprehensive evaluation of a bank's capacity to fulfill its debt obligations, influenced by both financial metrics and non-financial factors such as ESG performance. However, studies on the relationship between ESG ratings and credit ratings within the banking sector remain limited. Samaniego-Medina and Giraldez-Puig (2022) used ordered logistic regression to examine the effect of sustainability risk on the credit ratings of European banks between 2018 and 2022, finding that ESG controversies negatively affect credit ratings.

The growing emphasis on ESG performance in the banking sector makes it essential to further investigate its impact on credit ratings. While limited research directly addresses this relationship, existing studies suggest a significant link between ESG measures and credit ratings. This study aims to fill this gap by examining how different ESG dimensions influence banks' creditworthiness. Understanding this relationship will help banks better align their ESG strategies with financial goals and will enable credit rating agencies to incorporate ESG factors more effectively. These insights also provide valuable practical applications for banks, investors, and policymakers.

## 3. Hypotheses development

In the literature review, we introduced two contrasting theories on the impact of a company's ESG investments on its financial performance. Stakeholder theory (Freeman, 2023) suggests that a firm's long-term success hinges on its ability to manage relationships with all stakeholders, implying that ESG performance positively influences overall company performance. Conversely, agency theory (Jensen & Meckling, 1976) argues that ESG engagement may lead to conflicts between managers and shareholders, as ESG expenditures may not align with shareholder interests. Some scholars support this view, identifying a negative relationship between ESG performance and financial outcomes.

Due to the development of ESG regulations and current trend among credit rating agencies to integrate ESG factors in their rating methodologies, banks benefit from complying with ESG regulations and avoiding ESG-related risks. Therefore, we formulated our hypotheses in line with stakeholder theory. The following hypotheses were examined.

**Hypothesis 1 (H1).** ESG Combined Score (esg_com) positively impacts banks' credit ratings.

**Hypothesis 2 (H2).** Environmental Pillar Score (env) positively impacts banks' credit ratings.

**Hypothesis 3 (H3).** Social Pillar Score (soc) positively impacts banks' credit ratings.

**Hypothesis 4 (H4).** Corporate Governance Pillar Score (gov) positively impacts banks' credit ratings.

The previous study by Samaniego-Medina and Giraldez-Puig (2022) on European banks identified a negative relationship between ESG controversies and credit ratings. In our study, we used the ESG Controversies Score as a negative proxy for ESG controversies. This score ranges from 0 to 100 points, with 0 representing the worst performance and 100 the best, indicating the absence of controversies. Building upon the findings of the previous study, we tested the following hypothesis.

**Hypothesis 5 (H5).** ESG Controversies Score (esg_cont) positively impacts banks' credit ratings.

All three rating agencies—Moody's, S&P, and Fitch—have officially stated that they consider ESG factors as integral components of their rating processes. Therefore, despite potential differences in methodologies and the weightings assigned to various ESG factors, we expect that our hypotheses will hold true across all three agencies. Fig. 1 presents a conceptual model of our hypotheses (see Fig. 2).

Our study utilized ESG scores data extracted from Refinitiv (LSEG) database and our hypotheses were scientifically based on the ESG scoring methodology of LSEG.

LSEG categorizes a total of 186 ESG metrics into Environmental (E), Social (S), and Governance (G) categories, further divided into 10 sub-categories (Table 3). The overall ESG score is calculated as a weighted sum of the Environmental and Social scores, with the weights varying by industry, while the Governance weight remains consistent across all industries. The ESG Combined Score is determined by averaging the ESG Scores and the ESG Controversies Score for companies involved in ESG controversies. For companies without any controversies, the ESG Score equals the ESG Combined Score. In our study, we decomposed the ESG Combined Score into individual ESG Pillar Scores and the ESG Controversies Score, formulating hypotheses for each score (see Table 4).

## 4. Empirical analysis

### *4.1. Data sample*

We collected credit ratings, ESG scores and bank-specific financial performance data from the Refinitiv (LSEG) database. After excluding banks without ESG scores, our sample consists of 106 listed banks from 26 European countries. The credit ratings are from 2019 to 2023 to capture the latest data, while the ESG scores as well as financial performance scores are from 2018 to 2022 because of one year lag.

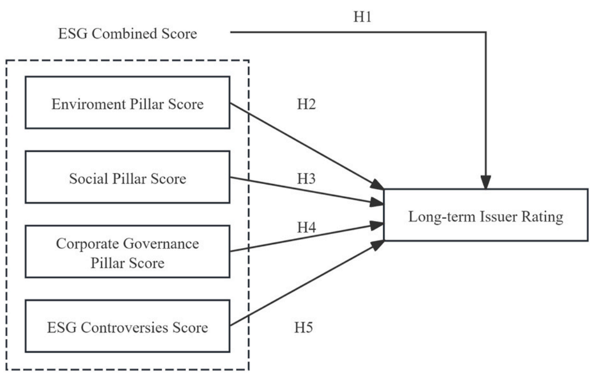


**Fig. 1.** Conceptual model of hypotheses.

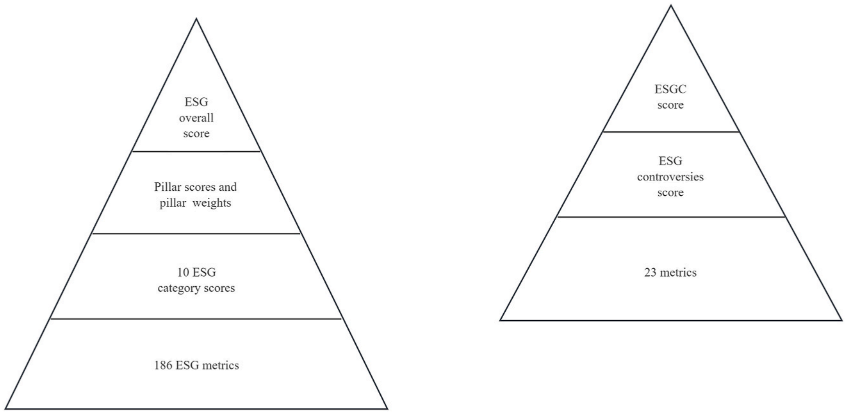


**Fig. 2.** Lsegesg scoring methodology.

**Table 3**
LSEGESG categories.

| ESG Pillars | ESG Categories |
|---|---|
| E | Resource use, Emissions, Innovation |
| S | Work force, Human rights, Community, Product responsibility |
| G | Management, Shareholders, CSR Strategy |

**Table 4**
Tabulation of countries.

| Country of headquarters | Frequency | Percent | Cum. |
|---|---|---|---|
| Austria | 17 | 3.16 % | 3.16 % |
| Belgium | 6 | 1.12 % | 4.28 % |
| Cyprus | 6 | 1.12 % | 5.39 % |
| Czech Republic | 11 | 2.04 % | 7.43 % |
| Denmark | 28 | 5.20 % | 12.64 % |
| Faroe Islands | 1 | 0.19 % | 12.83 % |
| Finland | 11 | 2.04 % | 14.87 % |
| France | 18 | 3.35 % | 18.22 % |
| Germany | 25 | 4.65 % | 22.86 % |
| Greece | 24 | 4.46 % | 27.32 % |
| Hungary | 6 | 1.12 % | 28.44 % |
| Iceland | 1 | 0.19 % | 28.62 % |
| Republic of Ireland | 18 | 3.35 % | 31.97 % |
| Italy | 72 | 13.38 % | 45.35 % |
| Liechtenstein | 8 | 1.49 % | 46.84 % |
| Netherlands | 12 | 2.23 % | 49.07 % |
| Norway | 30 | 5.58 % | 54.65 % |
| Poland | 48 | 8.92 % | 63.57 % |
| Portugal | 6 | 1.12 % | 64.68 % |
| Romania | 10 | 1.86 % | 66.54 % |
| Russia | 14 | 2.60 % | 69.14 % |
| Slovenia | 1 | 0.19 % | 69.33 % |
| Spain | 30 | 6.32 % | 75.65 % |
| Sweden | 20 | 3.72 % | 79.37 % |
| Switzerland | 35 | 6.51 % | 85.87 % |
| United Kingdom | 76 | 14.13 % | 100.00 % |
| **Total** | 538 | 100.00 % | |

## 4.2. *Description of variables*

### 4.2.1. *Dependent variables*

The dependent variables in this study are the long-term issuer credit ratings assigned to European listed banks by Moody's, S&P, and Fitch. These annually collected ratings evaluate the creditworthiness of banks over a long-term period (one year or more). The choice of ratings from these three agencies is due to their professional reputation and prominence in the credit rating industry.

In practice, it is common to observe multiple rating changes within a single year. To ensure consistency, we used the latest updated rating available for each bank at the end of the year. Since credit ratings are expressed in letter grades (e.g., AAA, AA+, etc.), we categorized them into 11 groups, scaled from 0 to 10. The lowest ranking (0) was assigned to banks with D ratings from S&P and Fitch or C ratings from Moody's, indicating default or near-default status. The highest ranking (10) was assigned to banks with ratings from AA + to AAA from S&P and Fitch or Aa2 to Aaa from Moody's, representing the highest credit quality.

The categorization method allows for a more straightforward quantitative analysis of the ratings data. Table 5 provides a detailed decomposition of the long-term issuer credit ratings assigned by Moody's, S&P, and Fitch.

#### 4.2.2. Independent variables

Independent variables comprised of ESG Combines Scores, Environmental Pillar Score, Social Pillar Score, Corporate Governance Pillar Score and ESG Controversies Score were obtained from Refinitiv database. Table 6 presents the description of independent variables (see Table 7).

#### 4.2.3. Control variables

The credit ratings of banks are significantly influenced by their financial performance. Strong financial performance can result in higher credit ratings, reflecting the bank's stability and lower risk of default, and vice versa.

Referring to Moody's rating methodology for banks (Moody's Investor Service, 2023), we consider solvency and liquidity as key indicators of a bank's financial performance. Solvency is measured through the non-performing loan (NPL) ratio, capital adequacy, and profitability ratios. These measures are consistent with previous studies, such as Liu et al. (2023) and Bruno et al. (2023), which examined the relationship between banks' non-performing loans and ESG performance. Capital adequacy, a proxy for liquidity risk and a bank's solvency to support its current capital structure, is commonly calculated as the ratio of tangible common equity to risk-weighted assets (RWA).

There are two primary approaches to measure financial performance: market-based and accounting-based (Van Beurden & Gössling, 2008). In this study, we focused solely on the accounting-based approach, using profitability ratios to measure bank performance. The key profitability ratios include Return on Assets (ROA), Return on Equity (ROE), and Net Interest Margin (NIM). We selected the ROA ratio as the indicator bank profitability because it is more commonly applied to banks. The ROA ratio was calculated by dividing net income by average tangible assets, which excludes goodwill and other intangible assets, to better determine a bank's ability to generate profits from its core operating assets.

Liquidity was assessed by the ratio of liquid assets to tangible assets, which measures a bank's ability to meet its short-term obligations using its liquid assets. According to the European Banking Authority (EBA) methodological guide, we also considered the debt-to-equity ratio and loan-to-deposits ratio as indicators of a bank's funding risk. These ratios were also included by Samaniego-Medina and Giraldez-Puig (2022) in their analysis of sustainability risks and credit ratings for European banks. As liquidity risk and funding risk are intertwined, Van den End (2014) found that the loan-to-deposit ratio is an effective tool in macroprudential policy to mitigate liquidity risk.

In addition, we considered size as a bank-specific variable, which is measured as the natural logarithm of the value of total assets in

**Table 5**
Decomposition of long-term issuer credit ratings of Moody's, S&P and Fitch.

| Long-Term Issuer Rating | | Fitch | Score | Ranking |
|---|---|---|---|---|
| Moody's | S&P | | | |
| Aaa Aa1 | AAA AA+ | AAA AA+ | 100 | 10 |
| | | | 95 | |
| Aa2 Aa3 | AA AA- | AA AA- | 90 | 9 |
| | | | 85 | |
| A1 A2 | A + A | A + A | 80 | 8 |
| | | | 75 | |
| A3 Baa 1 | A-BBB+ | A-BBB+ | 70 | 7 |
| | | | 65 | |
| Baa 2 Baa 3 | BBB BBB- | BBB BBB- | 60 | 6 |
| | | | 55 | |
| Ba1 Ba2 | BB + BB | BB + BB | 50 | 5 |
| | | | 45 | |
| Ba3 B1 | BB- B+ | BB- B+ | 40 | 4 |
| | | | 35 | |
| B2 B3 | B B- | B B- | 30 | 3 |
| | | | 25 | |
| Caa1 | CCC+ | CCC+ | 20 | |
| Caa2 | CCC | CCC | 20 | 2 |
| Caa3 | CCC- | CCC- | 20 | |
| Ca | CC C | CC C | 15 | 1 |
| | | | 10 | |
| C | RD | DDD | 5 | |
| / | SD | DD | 5 | 0 |
| / | D | D | 5 | |

**Table 6**
Description of independent variables.

| Variable | Description | Notation |
|---|---|---|
| ESG Combined Score | The ESG Combined Score provides a comprehensive assessment of a company's ESG performance based on information reported in the ESG pillars, with ESG controversies overlay captured from global media sources. | *esg_com* |
| Environmental Pillar Score | The Environmental Pillar Score is the weighted sum of the emission, resource use, and environmental innovation category scores. It measures a company's performance and impact in areas related to environmental sustainability. | *env* |
| Social Pillar Score | The Social Pillar Score is the weighted sum of the community, human rights, and product responsibility and workforce category scores. It measures a company's performance concerning its social responsibilities and relationships with stakeholders. | *soc* |
| Corporate Governance Pillar Score | The Corporate Governance Pillar Score is the weighted sum of the CSR strategy, management and shareholders category scores. It evaluates how effectively a company is governed and managed. | *gov* |
| ESG Controversies Score | The ESG Controversies Category Score measures a company's exposure to ESG controversies and negative events reflected in global media. The ESG Controversies Score ranges from 0 to 100 points, with 0 being the worst score and 100 being the best score, indicating no controversies. | *esg_cont* |

Source: ESG scores methodology from LSEG

**Table 7**
Description of control variables.

| Variable | Formula | Description | Notation |
|---|---|---|---|
| NPL ratio | Non-performing loans/ Gross loans | A higher NPL ratio indicates a higher risk of default and potential financial instability for the bank. | *nplr* |
| Capital adequacy | Tangible common equity /RWA | This ratio aligns with these regulatory requirements by focusing on the core, loss-absorbing capital relative to the risk of the bank's asset portfolio. | *cap_adq* |
| Profitability | Net income/Tangible assets | This ratio provides a clear and realistic view of a bank's ability to generate earnings from its tangible assets, excluding intangibles like goodwill that do not contribute directly to operational efficiency. | *prof* |
| Liquidity | Liquid assets/Tangible assets | The ratio of Liquid Assets to Tangible Assets directly measures a bank's readiness to meet its short-term obligations and cash flow needs. | *liq* |
| Debt-to- equity ratio | Total liabilities/Total equity | A high ratio indicates that a bank is heavily financed by debt, which could be risky if the bank faces financial difficulties. | *dte* |
| Loan to deposits | Total loans and advances/ Total deposits | A high ltd ratio indicates that the bank has lent out more than it has in deposits, which can be risky. A low ltd ratio suggests that the bank is not fully utilizing its deposit base for lending. | *ltd* |
| Bank size | n.a | Logarithm of total assets as a proxy of bank size | *size* |
| GDP growth | n.a | Annual real GDP growth to account for country-specific effect | *gdp_gr* |
| Country dummies | n.a | Country dummies to account for country-specific effect | *country dummy* |

EUR.

GDP growth was included as a country-specific control variable to account for the macroeconomic environment affecting banks. It influences financial performance by impacting loan demand, default rates, and overall economic conditions. Including GDP growth in our analysis helps isolate the impact of ESG performance on credit ratings from broader economic factors.

Lastly, we incorporated country dummies to control fixed effects at the country level. Country dummies are categorical variables that represent different countries in the dataset. Controlling for country-specific fixed effect enhances the robustness of our study by mitigating the potential bias from omitted variables or unobserved country-specific factors that could otherwise bias our results. The table below summarizes the control variables included in our study.

### *4.3. The mixed-effect ordered logistic regression model*

There are three main types of ordered logistic regression: random effects, fixed effects, and mixed effects. The random effects model assumes that individual differences are randomly distributed and uncorrelated with the predictors. In contrast, the fixed effects model controls for variables that vary across entities but remain constant over time. In our model, we included GDP growth and country dummies to account for country-level fixed effects. For this study, we adopted a mixed-effects ordered logistic model, which incorporates both fixed effects for predictors that are constant across entities and random effects to account for individual differences.

Following prior research (Chodnicka-Jaworska, 2021; Samaniego-Medina & Giraldez-Puig, 2022), we employed an ordered logistic panel data model to analyze the determinants of long-term issuer credit ratings assigned by Moody's, S&P, and Fitch.

The ordered logistic regression model is specified as:

$$Log\left[y_{i,j} \Big/ \left(1-y_{i,j}\right)\right] = \beta_0 + \beta_i X_{i,t-1} + \beta_j control\ variables_{i,t-1} + gdpgr_{i,t-1} + country\ dummy + \varepsilon_i$$

$y_{i,j}$ – rankings of long-term issuers ratings assigned by Moody's, S&P and Fitch.

$X_{i,t-1}$ – independent variables (esg_com, env, soc, gov, esg_cont)

Control variables$_{i,\ t\text{-}1}$ – nplr, cap_adq, prof, liq, dte, ltd, size
gdp_gr$_{i,t\text{-}1}$ – GDP growth
country dummy – country-level fixed effects.
$\varepsilon_i$ – idiosyncratic term.

Ordered logistic regression is used to model ordinal dependent variables, where the outcomes have natural order but the distances between categories are not assumed to be equal. In this study, the dependent variable $y_{i,j}$ represents the rankings of long-term issuer credit ratings, which are ordinal in nature. The model includes a range of independent variables $X_{i,t\text{-}1}$ and control variables that captured various aspects of a bank's performance and external environment as mentioned in Section 4.2. We also included country dummy variables to control for country-specific effects that could influence credit ratings, accounting for variations in economic, regulatory, and institutional environments across different countries.

We conducted a comprehensive analysis using six mixed-effects ordered logistic regressions to examine the impact of ESG performance on the credit ratings of European listed banks. The first three regressions (Reg. 1, Reg. 2, and Reg. 3) focus on the ESG Combined Score (esg_com) to evaluate the overall effect of ESG performance on bank credit ratings. The next three regressions (Reg. 4, Reg. 5, and Reg. 6) analyze the individual components of ESG, including the Environmental Pillar Score (env), Social Pillar Score (soc), Corporate Governance Pillar Score (gov), and the ESG Controversies Score (esg_cont).

As shown in Table 8, all our regression models are highly significant, with Prob > chi2 = 0.00, indicating strong predictive power of the independent variables on credit ratings. The Wald chi2 values range from 122.95 to 167.08, confirming the statistical significance of the models and the meaningful relationship between the predictors and the outcome variable. For example, Reg. 1, which examines the overall ESG Combined Score, has a Wald chi2 of 127.10, while Reg. 4, which includes individual ESG Pillar Scores and the ESG Controversies Score, has a Wald chi2 of 129.69. The Wald chi2 statistic is particularly valuable for confirming the robustness of the models. The consistent Wald chi2 results across different models suggest that our findings reflect a genuine relationship between ESG factors and credit ratings, rather than being influenced by model-specific anomalies.

In our model, we utilized lagged values of the independent and control variables to mitigate endogeneity issues and account for the fact that credit ratings are influenced by past financial performance and characteristics. This approach establishes a temporal relationship, demonstrating that prior ESG performance and financial metrics impact current credit ratings.

## 5. Results and discussions

### 5.1. Descriptive statistics

The descriptive statistics of the main variables are presented in Appendix 1. The long-term issuer credit ratings have a minimum ranking of 2 and a maximum ranking of 9, indicating that the highest credit ratings assigned to European listed banks by the three rating agencies range from AA-to AA, while the lowest ratings fall within the triple C range. Moody's displays a higher average rating of 7.081, suggesting that it tends to assign higher credit ratings to European listed banks compared to S&P and Fitch.

The results of ESG Combined Score, ESG Pillar Scores and ESG Controversies Score show significant variations. Medians show that most banks have ESG Combined Score above 55.8912 and ESG Pillar Scores above over 60. Additionally, a median of 100 of ESG Controversies Score indicates a high concentration of banks in the upper quartile of the scale.

We regressed the ESG Combined Score on the Environmental, Social, Corporate Governance Pillar Scores, and ESG Controversies Score to identify significant correlations. The results are presented in Appendix 2. The results indicate that the Environmental, Social, Governance, and ESG Controversies Scores are all significant positive predictors of the ESG Combined Score at the 1 % significance level. This finding aligns with the LSEG methodology, where the ESG Combined Score is calculated as a weighted sum of the individual ESG Pillar Scores and the ESG Controversies Score.

To address potential multicollinearity, we regressed banks' credit ratings on the ESG Combined Score, individual ESG Pillar Scores and ESG Controversies Score respectively to isolate its effect.

### 5.2. Panel regression results for the impact of ESG Combined Score on credit ratings

Table 9 summarizes the impact of the ESG Combined Score on long-term issuer credit ratings by Moody's, S&P, and Fitch. The results show that the ESG Combined Score does not have a statistically significant impact on the long-term issuer ratings of Moody's and S&P. However, it has a strong positive effect on the ratings assigned by Fitch. This finding aligns with Buallay's (2019) study, which identified a significant positive relationship between ESG and bank performance. Given the different results across the agencies, we can only partially accept Hypothesis 1, concluding that the ESG Combined Score positively impacts credit ratings assigned by Fitch.

Creditworthiness assesses an obligor's ability and willingness to meet financial obligations as they become due. Only ESG factors

**Table 8**
Model specifications for mixed-effect ordered logistic regressions.

| Specifications | Reg. 1 | Reg. 2 | Reg. 3 | Reg. 4 | Reg. 5 | Reg. 6 |
|---|---|---|---|---|---|---|
| Number of Obs. Wald chi2 | 264 | 250 | 266 | 264 | 250 | 266 |
| Prob > chi2 | 127.10 | 122.95 | 167.08 | 129.69 | 134.72 | 166.30 |
| | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |

**Table 9**
Comparative regression results - Impact of ESG Combined Score on long-term issuer credit ratings by Moody's, S&P, and Fitch.

| | Moody's | | S&P | | Fitch | |
|---|---|---|---|---|---|---|
| | Coefficient | P > z | Coefficient | P > z | Coefficient | P > z |
| *esg_com* | 0.010 | 0.227 | −0.006 | 0.510 | 0.032*** | 0.000 |
| *nplr* | −41.290*** | 0.000 | −42.616*** | 0.000 | −12.668*** | 0.000 |
| *cap_adq* | 9.457*** | 0.000 | 1.360 | 0.569 | 20.102*** | 0.000 |
| *prof* | 132.761*** | 0.000 | 49.190 | 0.122 | 103.881*** | 0.000 |
| *liq* | 0.179 | 0.897 | −0.489 | 0.742 | −4.373*** | 0.000 |
| *dte* | 0.205*** | 0.004 | 0.225** | 0.033 | 0.881*** | 0.000 |
| *ltd* | 0.135 | 0.783 | 0.488 | 0.391 | −3.360*** | 0.000 |
| *size* | 0.402*** | 0.000 | 0.600*** | 0.000 | 0.481*** | 0.000 |
| *gdp_gr* | −3.057 | 0.209 | −3.164 | 0.221 | −3.300 | 0.162 |
| *countrydummy* | −0.076*** | 0.000 | −0.031** | 0.044 | 0.004 | 0.804 |

Note: *, **, and *** denote significance at $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

that are relevant and material to creditworthiness are considered meaningful by rating agencies. Strong ESG performance does not necessarily equate to strong creditworthiness, and vice versa. The discrepancy in the impact of the ESG Combined Score on long-term issuer ratings across the three rating agencies may be explained by their differing methodologies. Moody's and S&P may evaluate E, S, and G factors as separate sources of credit risk rather than collectively when assessing a bank's credit profile. Although the Environmental, Social, and Governance factors contribute to the ESG Combined Score, its overall impact on credit ratings is less pronounced compared to individual Pillar Scores. This could be due to the dilution of each Pillar's contribution when calculating the weighted sum for the ESG Combined Score.

### *5.3. Panel regression results for the impact of ESG Pillar Scores and ESG Controversies Score on credit ratings*

Table 10 presents a comparative analysis of the regression results that examine the impact of Environmental, Social, and Governance Pillar Score as well as ESG Controversies Score on the long-term issuer credit ratings assigned by Moody's, S&P, and Fitch.

#### *5.3.1. The impact of environmental factors on banks' credit ratings*

We observed some consistency among the results, with all three credit rating agencies placing significant emphasis on environmental factors when assessing long-term issuer credit ratings. Hypothesis 2 is accepted for all three agencies. The Environmental Pillar Score has a significant positive impact on credit ratings, with coefficients significant at the 1 % level for Moody's and Fitch and at the 10 % level for S&P. This finding reflects the growing recognition of climate-related risks and their potential effect on financial stability. S&P classifies climate risk as one of the material risks in its credit rating methodology. Fitch also incorporates ESG factors when assessing the Viability Rating, which measures a financial institution's intrinsic creditworthiness. Although Moody's does not explicitly mention ESG factors in its methodology, the results suggest that environmental considerations may still be integrated into its rating process.

Our findings align with those of Jang et al. (2020) and Chodnicka-Jaworska (2021), who concluded that the Environmental Pillar Score is the most critical ESG factor in reducing companies' default risk. Additionally, Bauer and Hann (2010) observed that bond investors' awareness of environmental management practices has increased over the past decade. Their data show that companies neglecting environmental concerns tend to face higher debt financing costs and lower credit ratings. Conversely, companies that

**Table 10**
Comparative regression results – The Impact of ESG Pillar Scores and ESG Controversies Score on long-term issuer credit ratings by Moody's, S&P, and Fitch.

| | Moody's | | S&P | | Fitch | |
|---|---|---|---|---|---|---|
| | Coefficient | P > z | Coefficient | P > z | Coefficient | P > z |
| *env* | 0.021*** | 0.006 | 0.018* | 0.053 | 0.027*** | 0.000 |
| *soc* | −0.017 | 0.119 | −0.050*** | 0.000 | −0.009 | 0.441 |
| *gov* | 0.004 | 0.600 | −0.009 | 0.226 | 0.012* | 0.084 |
| *esg_cont* | 0.002 | 0.734 | 0.011* | 0.029 | 0.005 | 0.310 |
| *nplr* | −41.111*** | 0.000 | −48.045*** | 0.000 | −13.742*** | 0.000 |
| *cap_adq_* | 9.492*** | 0.000 | 2.641 | 0.277 | 21.152*** | 0.000 |
| *prof* | 125.093*** | 0.000 | 55.025* | 0.088 | 104.073*** | 0.000 |
| *liq* | −0.024 | 0.987 | 2.294 | 0.160 | −4.427*** | 0.001 |
| *dte* | 0.177** | 0.008 | 0.087 | 0.441 | 0.799*** | 0.000 |
| *ltd* | 0.156 | 0.744 | 0.989* | 0.096 | −3.180*** | 0.000 |
| *size* | 0.312** | 0.040 | 0.942*** | 0.000 | 0.331** | 0.021 |
| *gdp_gr* | −3.049 | 0.213 | −4.367* | 0.096 | −3.736 | 0.115 |
| *countrydummy* | −0.076** | 0.000 | −0.008 | 0.622 | 0.009 | 0.640 |

Note: *, **, and *** denote significance at$p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

proactively address environmental issues can reduce debt costs and potentially improve their credit scores.

The emphasis on environmental factors by rating agencies can be attributed to several reasons. First, climate-related risks, such as regulatory changes, physical climate risks, and the transition to a low-carbon economy are increasingly recognized as material financial risks that can affect a bank's asset quality, earnings, and capital adequacy. Second, growing investor demand for greater transparency and accountability on environmental issues is driving banks to improve their environmental performance. Third, regulatory bodies worldwide are tightening environmental regulations, which can impact banks' operations and financial stability, influencing their overall creditworthiness.

*5.3.2. The impact of social factors on banks' credit ratings*

The Social Pillar Score produced somewhat counterintuitive results in our analysis. All three credit rating agencies show a negative correlation between social factors and credit ratings, with S&P being the only one to exhibit a statistically significant relationship. S&P demonstrates a significant negative impact of the Social Pillar Score on credit ratings, while no significant effects were found for Moody's and Fitch. As a result, Hypothesis 3, which posited that the Social Pillar Score positively impacts banks' credit ratings, is rejected.

Several potential reasons could explain this phenomenon. First, social initiatives, such as improving labor practices, investing in community development, or enhancing customer relations often entail substantial upfront costs. These costs may negatively affect a company's financial performance in the short term, potentially leading to lower credit ratings. The long-term benefits, including increased customer loyalty, improved employee productivity, and enhanced corporate reputation, might not be immediately reflected in credit ratings.

Results from Buallay (2018) and Bătae et al. (2021) support this argument. Buallay found a negative relationship between corporate social responsibility (CSR) disclosure and three performance indicators when analyzing European banks. He suggests that executive management and boards of directors often develop CSR initiatives and pursue social policies for their own benefit, resulting in costs for banks that are ultimately borne by stakeholders. This can lower market value (Tobin's Q), equity (return on equity, ROE), and asset efficiency (return on assets, ROA). Similarly, Bătae et al. found that stakeholder theory's prediction of a positive relationship between corporate social responsibility and financial performance was not supported.

These perspectives align with our findings that the immediate financial implications of social initiatives can result in lower credit ratings, as rating agencies may prioritize short-term financial stability over potential long-term benefits.

Second, social metrics are often more challenging to quantify and standardize compared to environmental or governance metrics. The lack of consistent and comprehensive data can lead to variability in how social performance is assessed and reported. This inconsistency may result in a perceived negative impact on credit ratings, as rating agencies might adopt a conservative approach in the absence of reliable data.

*5.3.3. The impact of governance factors on banks' credit ratings*

The Corporate Governance Pillar Score presents mixed results across the three agencies. While Fitch shows a positive and statistically significant relationship between governance and credit ratings, no significant relationships are found for Moody's and S&P. Consequently, Hypothesis 4, which posits that the Corporate Governance Pillar Score (gov) positively impacts banks' credit ratings, is only partially supported. This inconsistency may stem from different methodological approaches or the weighting of governance factors in their credit rating models.

Like social factors, corporate governance factors are also challenging to measure, and some credit rating agencies incorporate these factors qualitatively. For example, S&P evaluates governance indicators such as governance and transparency, ownership structure, management quality, strategic positioning, operational effectiveness, financial management, and policies. The inherent uncertainty and complexity in assessing qualitative factors make it difficult to establish a direct connection between corporate governance quality and credit ratings, which may explain S&P's lack of a significant relationship.

Conversely, Fitch emphasizes quantitative governance factors, including weak senior management teams, high management turnover, over-dependence on key individuals, governance practices that threaten creditor interests, and low-quality or infrequent financial reporting or auditing. In a 2019 special report on the introduction of ESG Relevance Scores for financial institutions, Fitch indicated that Governance ESG risk elements significantly influenced ratings for both non-banking financial institutions (NBFIs) and banks. This focus may help explain the positive and significant correlation between corporate governance and credit ratings identified by Fitch.

Moody's, in contrast, emphasizes sovereign governance over corporate governance in its rating methodology. Governance is viewed as a key determinant of sovereign credit quality, and it is explicitly incorporated as a scorecard factor in Moody's sovereign rating approach. This omission likely contributes to the insignificant relationship between governance factors and bank credit ratings in Moody's model. The limited emphasis on corporate governance factors in Moody's methodology underscores the differences in results compared to Fitch.

Academic research supports the idea that strong governance can lead to improved credit outcomes. Studies by Zagorchev and Gao (2015) and Sohail et al. (2017) indicate that better governance is positively correlated with financial institution performance and negatively correlated with excessive risk-taking, aligning with Fitch's findings.

*5.3.4. The impact of ESG Controversies Score on banks' credit ratings*

The ESG Controversies Score, which ranges from 0 to 100 (with 0 indicating severe controversies and 100 indicating none), shows mixed results across the three credit rating agencies. While S&P observes a significant positive impact of fewer ESG controversies on credit ratings, Moody's and Fitch do not regard ESG controversies as critical in their assessments. Therefore, Hypothesis 5, positing that the ESG Controversies Score (esg_cont) positively impacts banks' credit ratings, is only partially supported.

Agencies may interpret the management and reporting of ESG controversies differently. A high ESG Controversies Score suggests effective governance and transparency practices, which could be favorably viewed by S&P. In its credit rating methodology for banks, S&P explicitly includes exposure to environmental, social, and governance-related risks as a subfactor of Risk Position. Such explicit consideration of ESG controversies is not found in the methodologies of Moody's and Fitch.

This discrepancy may be attributed to the differing methodologies and priorities of each agency. S&P's emphasis on risk management and transparency likely leads to a stronger focus on ESG controversies, viewing them as indicators of a bank's overall risk profile and governance quality. In contrast, Moody's and Fitch may incorporate ESG factors in a more holistic or indirect manner, potentially focusing more on other aspects of a bank's financial health and operational stability.

Samaniego-Medina & Pilar Giráldez-Puig (2022) conducted a study analyzing 65 European banks from 2011 to 2020, finding that lower levels of ESG controversies correlate with a higher probability of achieving the highest credit ratings. This finding aligns with our results for S&P, which shows a significant positive impact of fewer ESG controversies on credit ratings.

#### *5.3.5. The impact of financial indicators on banks' credit ratings*

In addition to ESG factors, our regression model includes several critical control variables influencing the long-term issuer credit ratings of banks. The non-performing loan ratio (NPLR) has a significant negative impact on credit ratings across all three rating agencies, underscoring the importance of asset quality in credit analysis. Capital adequacy (CA) is positively correlated with credit ratings for Moody's and Fitch, while profitability (PROF) positively and significantly impacts ratings across all agencies, suggesting that higher capital buffers and earnings capacity enhance creditworthiness. Liquidity (LIQ) shows mixed results, with a significant negative relationship between liquidity and credit ratings for Fitch. This counterintuitive finding may stem from Fitch's unique credit rating methodology. Table 11 in Section 5.4 indicates that Fitch uses the loan-to-deposit ratio to assess funding and liquidity, while our study measures liquidity using the ratio of liquid assets to tangible assets. This discrepancy suggests that Fitch's methodology may capture different aspects of liquidity risk, possibly leading to a negative relationship based on our liquidity measure. The debt-to-equity ratio (DTE) is positively correlated across all agencies, reflecting that higher leverage, within certain limits, may indicate market confidence. Higher leverage often signals better access to debt markets, which can be perceived as a sign of financial strength and stability. It may also suggest effective use of leverage to generate higher returns on equity, positively influencing credit ratings if banks are seen as capable of managing their debt responsibly. The loan-to-deposit ratio (LTD) presents varying impacts, with Fitch identifying a significant negative relationship. This finding may be justified by Fitch's focus on liquidity and funding stability in its credit rating methodology, as a higher loan-to-deposit ratio may indicate that a bank is overextended in lending relative to its deposit base,

**Table 11**
Comparison of methodologies for deriving standalone credit profiles of banks by S&P, Moody's, and Fitch.

| Aspect | S&P | Moody's | Fitch |
|---|---|---|---|
| Key Rating Metric | Standalone Credit Profile (SCP) | Baseline Credit Assessment (BCA) | Viability Rating (VR) |
| Macro Environment Analysis | Economic Risk, Industry Risk | Economic Strength, Institutions and Governance strength, Susceptibility to Event Risk | Operating Environment |
| Asset Quality | Risk appetite, Loss experience and expectations etc | Non-Performing Loans (NPL) Ratio | Impaired Loans Ratio |
| Capital Adequacy | Total Adjusted Capital (TAC), Regulatory Capital Metrics | Tangible common equity/RWA | Core capital ratio |
| Earnings and Profitability | Earnings capacity and quality | Return on Assets (ROA), Net Profit Margin | Operating Profit/RWAs |
| Funding and Liquidity | Wholesale sale funding base, deposit stability, access to funding etc | Market Funds/Tangible Banking Assets, Liquid Banking Assets/Tangible Banking Assets | Loan-to-Deposit Ratio |
| Qualitative Adjustments | Business Position | Business Diversification, Opacity and Complexity, Corporate Behavior | Business Profile, Risk Profile |
| Financial Focus | Not Clear | Asset Risk and Capital Quality | Capitalization and Leverage |
| ESG Considerations | Climate risk is included in other material risks in the Risk Position Assessment. Governance is considered as subfactor in the Business Position Assessment. Exposure to environmental, social, and governance-related risks describe the subfactor of Risk Position | Not mentioned | ESG factors have a significant influence on a bank's VR, this is likely to be via the Business Profile or Risk Profile Key Rating Drivers. |

**Table 12**
Result of robustness test for Moody's.

| Variable | Coefficient | P > z | Variables | Coefficient | P > z |
|---|---|---|---|---|---|
| *env* | 0.021*** | 0.006 | *esgs* | 0.007 | 0.440 |
| *soc* | −0.017 | 0.119 | | | |
| *gov* | 0.004 | 0.600 | | | |
| *esg_cont* | 0.002 | 0.734 | *esg_cont* | 0.003 | 0.528 |
| *nplr* | −41.111*** | 0.000 | *nplr* | −41.551*** | 0.000 |
| *cap_adq_* | 9.492*** | 0.000 | *cap_adq_* | 9.316*** | 0.000 |
| *prof* | 125.093*** | 0.000 | *prof* | 132.350*** | 0.000 |
| *liq* | −0.024 | 0.987 | *liq* | 0.231 | 0.873 |
| *dte* | 0.177** | 0.008 | *dte* | 0.200*** | 0.005 |
| *ltd* | 0.156 | 0.744 | *ltd* | 0.195 | 0.690 |
| *size* | 0.312** | 0.040 | *size* | 0.401*** | 0.005 |
| *gdp_gr* | −3.049 | 0.213 | *gdp_gr* | −3.051 | 0.211 |
| *countrydummy* | −0.076*** | 0.000 | *countrydummy* | −0.076*** | 0.000 |

Note. *, **, and *** denote significance at $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

**Table 13**
Result of robustness test for S&P.

| Variables | Coefficient | P > z | Variables | Coefficient | P > z |
|---|---|---|---|---|---|
| *env* | 0.018* | 0.053 | *esgs* | −0.039*** | 0.000 |
| *soc* | −0.050*** | 0.000 | | | |
| *gov* | −0.009 | 0.226 | | | |
| *esg_cont* | 0.011* | 0.029 | *esg_cont* | 0.012** | 0.016 |
| *nplr* | −48.045*** | 0.000 | *nplr* | −46.455*** | 0.000 |
| *cap_adq_* | 2.641 | 0.277 | *cap_adq_* | 1.946 | 0.418 |
| *prof* | 55.025* | 0.088 | *prof* | 65.011** | 0.046 |
| *liq* | 2.294 | 0.160 | *liq* | 1.793 | 0.265 |
| *dte* | 0.087 | 0.441 | *dte* | 0.209** | 0.048 |
| *ltd* | 0.989* | 0.096 | *ltd* | 0.584 | 0.302 |
| *size* | 0.942*** | 0.000 | *size* | 1.057*** | 0.000 |
| *gdp_gr* | −4.367* | 0.096 | *gdp_gr* | −4.387* | 0.095 |
| *countrydummy* | −0.008 | 0.622 | *countrydummy* | −0.008 | 0.605 |

Note. *, **, and *** denote significance at $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

**Table 14**
Result of robustness test for Fitch.

| Variables | Coefficient | P > z | Variables | Coefficient | P > z |
|---|---|---|---|---|---|
| *env* | 0.027*** | 0.000 | *esgs* | 0.026*** | 0.003 |
| *soc* | −0.009 | 0.441 | | | |
| *gov* | 0.012* | 0.084 | | | |
| *esg_cont* | 0.005 | 0.310 | *esg_cont* | 0.006 | 0.223 |
| *nplr* | −13.742*** | 0.000 | *nplr* | −11.875*** | 0.000 |
| *cap_adq_* | 21.152*** | 0.000 | *cap_adq_* | 19.927*** | 0.000 |
| *prof* | 104.073*** | 0.000 | *prof* | 110.955*** | 0.000 |
| *liq* | −4.427*** | 0.001 | *liq* | −4.210*** | 0.001 |
| *dte* | 0.799*** | 0.000 | *dte* | 0.863*** | 0.000 |
| *ltd* | −3.180*** | 0.000 | *ltd* | −3.159*** | 0.000 |
| *size* | 0.331** | 0.021 | *size* | 0.411*** | 0.002 |
| *gdp_gr* | −3.736 | 0.115 | *gdp_gr* | −3.128 | 0.184 |
| *countrydummy* | 0.009 | 0.640 | *countrydummy* | −0.001 | 0.948 |

Note. *, **, and *** denote significance at $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

raising concerns about liquidity risks and the ability to meet short-term obligations. The size of the bank consistently shows a positive relationship with credit ratings, implying that larger banks are perceived as more stable and influential. Lastly, macroeconomic variables like GDP growth rate (GDPGR) and country-specific factors (country dummy) demonstrate inconsistent significance, reflecting differing emphases on these factors in each agency's rating methodology (see Tables 12–14).

### *5.4. Comparison of credit rating methodologies for banks across rating agencies*

The assessment of banks' creditworthiness is a multifaceted process that involves various quantitative and qualitative factors. S&P, Moody's, and Fitch employ distinct methodologies to evaluate the standalone credit profile (SCP) of banks, which reflects a bank's intrinsic financial strength, independent of potential external support from governments or affiliated entities. Comparing the similarities and differences in assessing the SCP helps us understand the results of our regression model.

The methodologies of S&P, Moody's, and Fitch for deriving the standalone credit profiles of banks share several commonalities. All three agencies conduct comprehensive assessments that include both quantitative and qualitative factors. They evaluate the macroeconomic environment, business position, capital adequacy, earnings and profitability, asset quality, risk management, and funding and liquidity. Additionally, each agency incorporates an analysis of qualitative factors, such as geographical diversification and business competitiveness. The evaluation process is rigorous, involving committee reviews and ongoing monitoring to ensure that the ratings remain current and reflective of the banks' financial health. These methodologies aim to provide a holistic view of a bank's creditworthiness, balancing financial metrics with qualitative assessments to capture the full spectrum of risks and strengths.

However, notable differences exist in the rating approaches used by S&P, Moody's, and Fitch. Moody's relies on five key financial ratios as the foundation for its rating implications, which include capital adequacy, asset quality, profitability, and funding and liquidity. It then adjusts these implied ratings by considering additional sub-factors, either quantitatively or qualitatively, to derive the financial profile. Based on this financial profile, Moody's incorporates three qualitative adjustment factors to determine the standalone credit profile.

Fitch employs a similar approach to Moody's but assigns different financial ratios and weightings. For instance, Moody's allocates a 35 % weighting to funding and liquidity factors, while Fitch assigns only 10 %. Instead, Fitch emphasizes aspects such as capital ratios and leverage ratios. Additionally, Fitch's methodology includes a detailed examination of the business and risk profiles, ensuring that these qualitative factors are integrated into the final rating, with ESG factors also mentioned.

In contrast, S&P follows a slightly different methodology. S&P does not begin with a standard set of financial ratios to derive an implied rating or assign specific weightings to sub-factors. Instead, its approach is more holistic, combining both qualitative and quantitative analyses from the outset. S&P evaluates the bank's business position, capital and earnings, risk position, funding, and liquidity, but the final rating is determined through a committee process that considers the interplay of these factors without predefined weightings.

In summary, while S&P, Moody's, and Fitch all conduct thorough assessments of banks' financial and non-financial factors, their methodologies differ in terms of initial focus on financial ratios, weightings assigned to various factors, and integration of qualitative adjustments. These differences can lead to variations in the final credit ratings assigned to the same institution by different agencies.

The differences in rating methodologies help explain the variability in the significance and impact of control variables in the regression model. For example, the significance of capital adequacy for Moody's and Fitch aligns with their emphasis on financial ratios. The mixed results for liquidity can be attributed to S&P's qualitative approach, which may not isolate liquidity as a standalone critical factor. Similarly, the varying significance of long-term debt reflects different tolerances and considerations for long-term financial commitments.

Overall, understanding the methodologies of S&P, Moody's, and Fitch provides a framework for interpreting the impact of control variables on credit ratings. These differences influence how factors such as NPLR, capital adequacy, and profitability affect the long-term issuer credit ratings of banks.

### *5.5. Robustness test*

As discussed in Section 5.2, LSEG calculates the ESG overall score (esgs) as a weighted sum of Environmental, Social and Governance category scores. To ensure the validity and reliability of our results, we conducted a robustness test by replacing E, S and G Pillar Scores by the ESG overall score (esgs). We replicated the previous analysis by running a regression of credit rating scores on the ESG overall score and ESG Controversies Score, while keeping the control variables unchanged. The tables below present the results of robustness test for Moody's, S&P and Fitch.

In our previous analysis, the Environmental Score had a positive impact on Moody's credit ratings at the 1 % significance level. However, the ESG overall score loses its significance in the robustness test. This discrepancy may be attributed to Moody's asymmetric E, S, and G issuer scoring scale, which suggests that ESG issues are more often viewed as sources of credit risk rather than strengths, particularly concerning environmental and social issues. Consequently, environmental factors may act as the primary source of credit risk, rather than a joint influence from environmental, social, and governance factors on the credit profiles of banks.

The effects of the ESG Controversies Score and the control variables remain consistent with the results from the previous study.

The results of the robustness test for S&P align with the findings from the previous study. The small p-value for the ESG overall score indicates a significant negative effect on credit ratings at the 1 % significance level, primarily driven by the negative impact of the Social Pillar on credit ratings. Additionally, the effect of the ESG Controversies Score is reinforced, with a p-value of 0.016 indicating a positive impact on credit ratings at the 5 % level.

The results of the robustness test for Fitch further reinforce the findings of our previous study. The p-value of 0.003 indicates that the ESG overall score has a positive and significant impact on Fitch's credit ratings at the 1 % significance level, primarily driven by the positive and significant effects of the Environmental and Governance Pillars.

In summary, the robustness test results for Moody's highlight a difference between the impacts of individual ESG Pillar Scores and the ESG overall score, which may be attributed to Moody's asymmetric ESG issuer scoring scale. In contrast, our results remain robust for S&P and Fitch.

## 6. Conclusions and implications for credit rating agencies

This paper aimed to examine the impact of Environmental, Social, and Governance (ESG) factors on the long-term issuer credit ratings of European listed banks as assigned by Moody's, S&P, and Fitch. Using ordered logistic regression models, we assessed the significance and direction of relationships between various ESG metrics and credit ratings while controlling traditional financial and macroeconomic variables.

Our analysis reveals several key findings. The ESG Combined Score shows a significant and positive relationship only in Fitch's ratings, indicating Fitch's greater emphasis on aggregated ESG performance. The Environmental Pillar Score consistently demonstrates a positive and significant impact on credit ratings across Moody's, S&P, and Fitch, underscoring the importance of strong environmental practices in assessing creditworthiness. Conversely, the Social Pillar Score yields mixed results: a significant negative correlation is found with S&P, while no significant relationships are observed with Moody's and Fitch, suggesting that the short-term costs of social initiatives might outweigh their immediate benefits. The Corporate Governance Pillar Score positively and significantly impacts Fitch's credit ratings but not those of Moody's and S&P. This discrepancy is likely due to differences in methodologies and the emphasis on qualitative versus quantitative factors. The ESG Controversies Score shows a significant positive impact on S&P's credit ratings, highlighting S&P's focus on transparency and risk management; however, controversy factors are not deemed critical by Moody's and Fitch. Additionally, control variables such as profitability, capital adequacy, and bank size positively influence credit ratings, while the non-performing loan ratio shows negative associations.

From a theoretical standpoint, we can confidently affirm the stakeholder theory based on the consistent and significantly positive results of environmental factors on credit ratings. This finding bolsters the arguments made by scholars such as Jang et al. (2020) and Chodnicka-Jaworska (2021). However, while the ESG Combined Score, Corporate Governance Score, and ESG Controversies Score show positive correlations with credit ratings, these relationships are not consistently significant across different credit rating agencies. Therefore, we cannot fully support the stakeholder theory in these respects. Additionally, all three credit rating agencies exhibit a negative correlation—whether significant or not—between the Social Pillar Score and credit ratings. This aligns with analyses by Buallay (2018) and Bătae et al. (2021), suggesting that the short-term costs associated with social initiatives may outweigh their immediate benefits. This evidence provides partial support for agency theory, which posits that management decisions may not always align with shareholders' interests, particularly when significant social investments are involved.

This paper highlights the profound influence of environmental factors on credit assessments for European banks, highlighting the financial sector's growing recognition of climate risks and their potential systemic impact on the global economy. While environmental factors have been identified as influential in credit assessments, credit rating agencies can improve their methodologies by adopting standardized frameworks such as the Science-Based Targets Initiative (SBTi) or the Task Force on Climate-related Financial Disclosures (TCFD), which would enhance transparency, comparability, and alignment with global best practices, enabling better-informed investment decisions.

Social and governance factors exhibit more complex relationships with credit ratings. Unlike environmental performance, which benefits from established frameworks and guidelines like the SBTi for calculating carbon emissions, social and governance aspects are inherently more difficult to quantify. Social performance, covering a wide range of issues, including labor practices, community impact, and human rights, is challenging to quantify and standardize, leading to inconsistent assessments and reporting. Similarly, governance performance, involving corporate structures, management quality, and board effectiveness, often requires subjective judgment due to lack of standardized metrics. These challenges result in varying weighting and interpretations across different rating agencies. Without standardized measures, the impact of social and governance factors on credit ratings remains unclear, highlighting the necessity for transparent ESG reporting and evaluation frameworks.

Notably, our findings reveal a negative correlation between social factors and credit ratings across all three agencies, suggesting that the short-term costs associated with promoting social initiatives may outweigh their immediate benefits in the eyes of credit assessors. This trade-off arises as companies investing in areas like labor practices, community development, or customer relations often face significant upfront costs, impacting short-term financial performance and credit ratings. While such initiatives can yield long-term benefits like improved loyalty, productivity, and reputation, these outcomes may not align with the short-to medium-term financial metrics prioritized in credit evaluations. Credit rating agencies should also consider how social investments foster long-term sustainability and stakeholder trust, despite short-term financial impacts when integrating social factors into rating methodologies.

This study has several limitations. First, its focus on a specific timeframe and European banks may restrict the generalizability of findings to other regions or financial institutions. Changes in economic conditions, regulations, and market trends occurring after the study period could affect the relationships between ESG factors and credit ratings. Second, while correlations between ESG factors and credit ratings are analyzed, establishing causality is challenging due to potential unobserved variables or external influences. Third, the qualitative nature of ESG assessments introduces subjectivity and potential biases, which may affect result consistency. Fourth, ESG data constraints limited the sample to 106 European listed banks, potentially skewing toward larger institutions and resulting in an unbalanced dataset due to inconsistent reporting over the five-year period. Finally, the use of credit ratings from Moody's, S&P, and Fitch, each with distinct methodologies, may introduce variability in control variable outcomes despite the comparative analysis.

Future research could build on our findings by addressing key limitations and exploring new avenues. First, expanding the analysis to include a broader range of financial institutions beyond European banks, such as those from different regions and sizes, would offer a more global perspective on ESG-credit rating relationships. Second, investigating causal mechanisms through qualitative methods could clarify how ESG initiatives impact financial outcomes and influence investor and rating agency perceptions.

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## CRediT authorship contribution statement

Wang Danni: Writing – review & editing, Writing – original draft, Formal analysis, Project administration, Conceptualization, Methodology. Chen Shengli: Writing – review & editing, Methodology.Chen Haowei:Writing – review & editing, Software.

## Appendix

*Appendix 1. Descriptive statistics of the main variables*

| Variable | Obs | Mean | Std. dev. | Min | Max |
|---|---|---|---|---|---|
| Moody's | 284 | 7.081 | 1.467 | 2.000 | 9.000 |
| S&P | 277 | 6.682 | 1.496 | 2.000 | 9.000 |
| Fitch | 293 | 6.604 | 1.497 | 2.000 | 9.000 |
| esg_com | 446 | 55.771 | 15.707 | 1.397 | 88.472 |
| env | 446 | 60.280 | 28.163 | 0.000 | 98.108 |
| soc | 446 | 63.969 | 19.236 | 1.217 | 97.640 |
| gov | 446 | 59.545 | 21.820 | 2.205 | 95.134 |
| esg_cont | 446 | 81.912 | 30.878 | 0.500 | 100.000 |
| nplr | 445 | 0.050 | 0.120 | 0.000 | 1.600 |
| cap_adq | 425 | 0.171 | 0.062 | −0.008 | 0.455 |
| prof | 446 | −0.016 | 0.492 | −10.373 | 0.041 |
| liq | 446 | 0.190 | 0.120 | 0.000 | 0.675 |
| dte | 446 | 2.316 | 2.650 | 0.000 | 28.108 |
| ltd | 438 | 0.981 | 0.409 | 0.000 | 3.352 |
| TotalAssets | 446 | 298222 | 516445 | 0.000 | 2755813 |
| GDP | 446 | 0.021 | 0.049 | −0.112 | 0.151 |
| Country dummy | 538 | 16.119 | 7.479 | 1 | 26 |

*Appendix 2. Result of multicollinearity test*

| *esg_com* | Coefficient | Std. errs. | z | P > z |
|---|---|---|---|---|
| *env* | 0.111*** | 0.012 | 9.130 | 0.000 |
| *soc* | 0.511*** | 0.019 | 26.450 | 0.000 |
| *gov* | 0.345*** | 0.012 | 28.420 | 0.000 |
| *esg_cont* | 0.310*** | 0.009 | 35.140 | 0.000 |
| *nplr* | −1.255 | 1.944 | −0.650 | 0.519 |
| *cap_adq* | 3.761 | 3.984 | 0.940 | 0.345 |
| *prof* | −68.179* | 38.172 | −1.790 | 0.074 |
| *liq* | −1.175 | 2.355 | −0.500 | 0.618 |
| *dte* | −0.021 | 0.121 | −0.170 | 0.863 |
| *ltd* | 1.438* | 0.862 | 1.670 | 0.095 |
| *size* | −0.325 | 0.251 | −1.290 | 0.196 |
| *gdp_gr* | 6.381 | 3.957 | 1.610 | 0.107 |
| *country_dummy* | −0.017 | 0.035 | −0.500 | 0.621 |
| _cons | −27.083*** | 3.209 | −8.440 | 0.000 |
| sigma_u | 1.553 | | | |
| sigma_e | 3.210 | | | |
| rho | 0.190 | | | |

Note. *, **, and *** denote significance at$p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

## Data availability

Data will be made available on request.

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